EE596H: Quiz

Total Questions: 2, Total Marks: 25, Time: 90 Mins

1. For each of the following matrices determine the values of α for which they are positive definite / negative definite / positive semidefinite / negative semidefinite / indefinite:

   (a) $A_\alpha = \begin{pmatrix} \alpha & 1 \\ 1 & \alpha \end{pmatrix}$ $\boxed{5}$

   (b) $B_\alpha = \begin{pmatrix} \alpha & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$ $\boxed{5}$

   (c) $C_\alpha = \begin{pmatrix} \alpha & 0 & 1 \\ 0 & \alpha & 0 \\ 1 & 0 & \alpha \end{pmatrix}$ $\boxed{5}$

   Total: 15

2. Consider a function $f : [0,1]^n \to \mathbb{R}$ given by

$$f(\mathbf{x}) = -\sqrt{-\sum_{j=1}^{n} x_j \log(x_j)}$$

Which one of the following is true? Prove in details.

1. $f(\mathbf{x})$ is convex.
2. $f(\mathbf{x})$ is concave.
3. Neither of the above two are true in general.

$\boxed{10}$

4.4
0.9
5 3

$-\sqrt{-x_1\log x_1 - x_2\log x_2}$

$-x_1 \log x_1$

$x_1\log x_1 + x_2\log x_2$

$x_1 \times \frac{1}{x_1} + \log x_1$

$1 + \log x_1$

0

$\frac{1}{x_1}$ 0 $\frac{1}{x_1 x_2}$

0 $\frac{1}{x_2}$